MTH 261 **4/23/2026**
Spring 2026 **Midterm Exam #1** **1 Hour 40 Minutes**

Name (Print): _____

This exam contains 11 pages (including this cover page) and 9 problems. Check to see if any pages are missing. Print your name above.

You are allowed one sheet of notes. Turn in your notesheet along with your response. You are also allowed to use a calculator on the exam.

Read the following instructions carefully:

- You can only use methods discussed in class so far. i.e., you may **NOT** use any method we haven't learned in class so far.
- **Show ALL your work!** You are being assessed on your work and explanations. Correct answers with little to no work or explanation will not receive full credit. I give partial credits so don't leave a blank if you have some ideas.
- **Organize your work.** Your work should be clearly written and organized. Please use correct mathematical notations when appropriate. For example, please use vector notations when working with vectors and don't use vector notations when working with scalars and points.
- By submitting this exam, you agree that all work represents your individual effort and understanding. Failure to comply with any of the above statements will be considered a violation of the Student Code of Conduct and will result in a 0 for this assessment and possible disciplinary action.

Problem	Points	Score
1	12	
2	12	
3	13	
4	12	
5	12	
6	12	
7	13	
8	14	
9	0	
Total:	100	

Do not write in the table to the right. Adding the bonus, the total score won't exceed 100%.

Please keep in mind that an assessment is just another means of communication between you and me to help me understand what you have learned so far. It is not a measure of your worth as a person nor your intelligence or aptitude as a student. Just give this your best try.

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1. (12 points) Solve the following system of linear equations using the Gaussian Elimination algorithm:

$$\begin{cases} 5x + 7y - 2z = 3 \\ x + 2y - z = 1 \\ -6y + 6z = 2 \end{cases}$$

If the system is consistent, write the solution set as a column vector. If the system is inconsistent, explain how you know it.

2. (12 points) Solve the following system of linear equations using the Gaussian Elimination algorithm:

$$\begin{cases} x + y + z = 8 \\ 2x - y + z = 9 \\ x + 2y - z = 1 \end{cases}$$

If the system is consistent, write the solution set as a column vector. If the system is inconsistent, explain how you know it.

3. (13 points) Let $A = \begin{bmatrix} 1 & -2 & 1 & -3 \\ 2 & -4 & 4 & 2 \\ -1 & 2 & -3 & 1 \\ 3 & -6 & 3 & 9 \end{bmatrix}$, describe all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$ in parametric form.

4. (12 points) Let $A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 2 \\ -1 & 3 \end{bmatrix}$, and $C = \begin{bmatrix} 2 & -1 \\ 0 & 4 \\ 1 & 5 \end{bmatrix}$. Compute the following expressions if possible. If not possible, explain why.

   (a) $2A - 3C^T$

   (b) $(A + 2B)^T$

5. (12 points) Let $A = \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 4 & -1 \\ -3 & 2 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 3 & 2 \end{bmatrix}$. Multiply the matrices as indicated if possible. If not possible, explain why.

(a) AB

(b) BA

6. (12 points) Is the following matrix invertible? If so, find its inverse. If not, explain why the inverse doesn't exist.

   (a) $A = \begin{bmatrix} 15 & -12 \\ -20 & 16 \end{bmatrix}$

   (b) $B = \begin{bmatrix} 17 & -12 \\ -10 & 7 \end{bmatrix}$

7. (13 points) Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & -2 & -1 \end{bmatrix}$.

(a) Find A^{-1}, the inverse of the matrix A.

(b) Solve the matrix equation $A\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ using A^{-1} you found in part (a).

8. (14 points) Let $A = \begin{bmatrix} 0 & 3 & 0 \\ 1 & 2 & 0 \\ 4 & 8 & 1 \end{bmatrix}$. Express the matrix A **and** its inverse A^{-1} as a product of elementary matrices.

9. (10 points (bonus)) **This is a BONUS problem.**

Suppose experimental data are represented by a set of points in the plane. An **interpolating polynomial** for the data is a polynomial whose graph passes through every point. In scientific work, such a polynomial can be used, for example, to estimate values between the known data points. Another use is to create curves for graphical images on a computer screen.

One method for finding an interpolating polynomial is to solve a system of linear equations.

Find the interpolating polynomial $p(t) = a_0 + a_1 t + a_2 t^2$ for the data $(1, 12)$, $(2, 15)$, $(3, 16)$. That is, find the values of a_0, a_1, and a_2 for this polynomial such that the graph of this polynomial will pass through these three points.